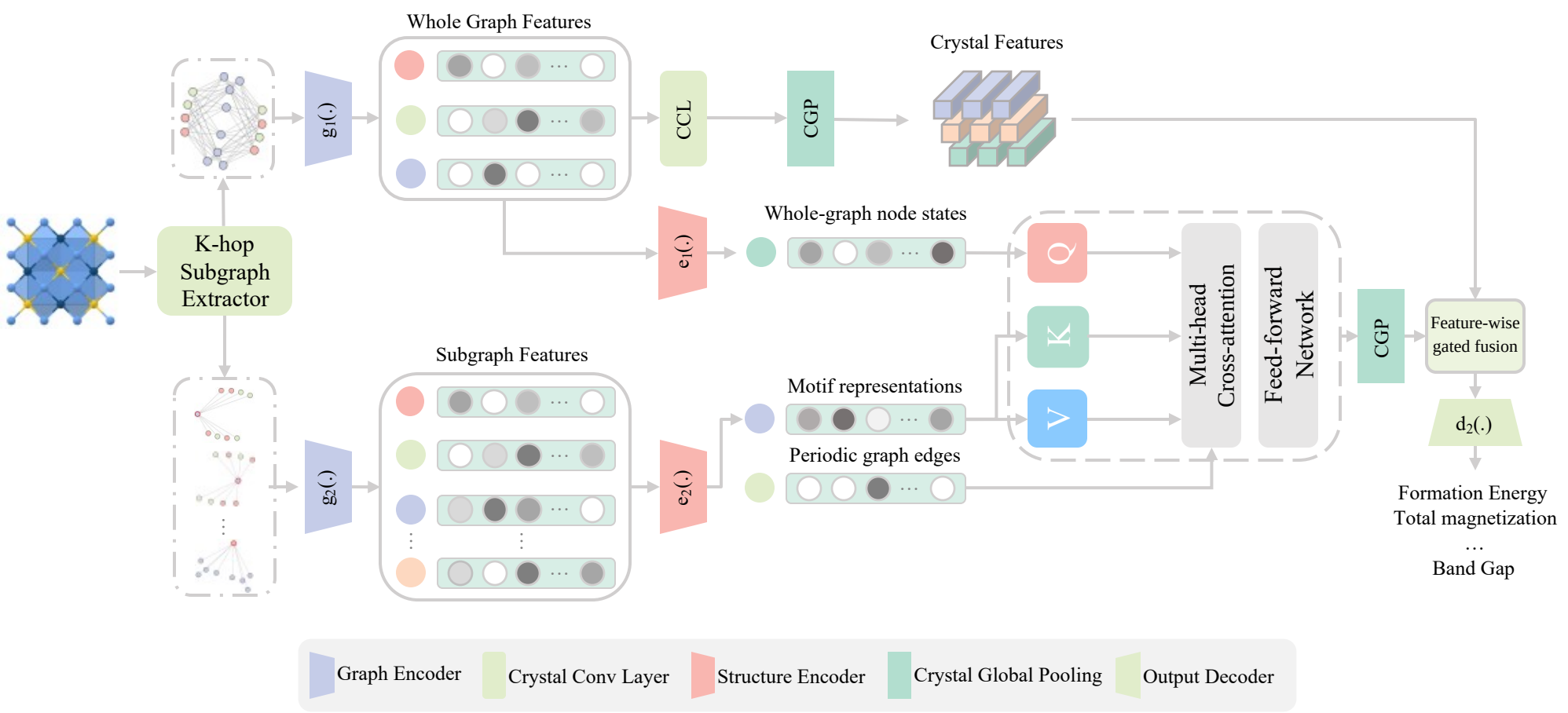

Whole Graph Features
Crystal Features
$g_1(.)$
CCL
CGP
K-hop Subgraph Extractor
$e_1(.)$
Whole-graph node states
Q
K
V
Multi-head Cross-attention
Feed-forward Network
CGP
Feature-wise gated fusion
$d_2(.)$
Formation Energy
Total magnetization
…
Band Gap
Subgraph Features
$g_2(.)$
$e_2(.)$
Motif representations
Periodic graph edges
Graph Encoder
Crystal Conv Layer
Structure Encoder
Crystal Global Pooling
Output Decoder